

Wrapper Class

Q70. Write a Java program that generates the minimum and maximum value for each of the Numeric Wrapper classes (Byte, Short, Integer, Long, Float, Double).

Code:

```
Module 4 > J WrapperRange.java > WrapperRange
1 public class WrapperRange {
2
3     Run | Debug
4     public static void main(String[] args) {
5
6         System.out.println(x: "Integer range:");
7         System.out.println("min: " + Integer.MIN_VALUE);
8         System.out.println("max: " + Integer.MAX_VALUE);
9
10        System.out.println(x: "\nDouble range:");
11        System.out.println("min: " + Double.MIN_VALUE);
12        System.out.println("max: " + Double.MAX_VALUE);
13
14        System.out.println(x: "\nLong range:");
15        System.out.println("min: " + Long.MIN_VALUE);
16        System.out.println("max: " + Long.MAX_VALUE);
17
18        System.out.println(x: "\nShort range:");
19        System.out.println("min: " + Short.MIN_VALUE);
20        System.out.println("max: " + Short.MAX_VALUE);
21
22        System.out.println(x: "\nByte range:");
23        System.out.println("min: " + Byte.MIN_VALUE);
24        System.out.println("max: " + Byte.MAX_VALUE);
25
26        System.out.println(x: "\nFloat range:");
27        System.out.println("min: " + Float.MIN_VALUE);
28        System.out.println("max: " + Float.MAX_VALUE);
29    }
```

Output:

```
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PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes> cd "C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4"
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> javac WrapperRange.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> java WrapperRange

Integer range:
min: -2147483648
max: 2147483647

Double range:
min: 4.9E-324
max: 1.7976931348623157E308

Long range:
min: -9223372036854775808
max: 9223372036854775807

Short range:
min: -32768
max: 32767

Byte range:
min: -128
max: 127

Float range:
min: 1.4E-45
max: 3.4028235E38
PS C:\Users\ArkaLaptop\Documents\VScode Codes\
```

Q71. Write a Java program to receive an integer number as a command line argument, and print the binary, octal, and hexadecimal equivalent of the given number.

```
Module 4 > J NumberConversion.java > NumberConversion
1 public class NumberConversion {
2
3     Run | Debug
4     public static void main(String[] args) {
5
6         if (args.length != 1) {
7             System.out.println(x: "Usage: java NumberConversion <number>");
8             return;
9         }
10
11         int number = Integer.parseInt(args[0]);
12
13         System.out.println("Given Number : " + number);
14         System.out.println("Binary equivalent : " + Integer.toBinaryString(number));
15         System.out.println("Octal equivalent : " + Integer.toOctalString(number));
16         System.out.println("Hexadecimal equivalent : " + Integer.toHexString(number));
17     }
18 }
```

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```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> javac NumberConversion.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> java NumberConversion 20
Given Number : 20
Binary equivalent : 10100
Octal equivalent : 24
Hexadecimal equivalent : 14
```

Q72. Write a Java program that reads an integer number (between 1 and 255) from the user and prints the binary representation of the number as an 8-bit string. If the binary string has fewer than 8 digits, pad it with leading zeros.

```
Module 4 > J BinaryRepresentation.java > BinaryRepresentation
1  import java.util.Scanner;
2
3  public class BinaryRepresentation {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          Scanner sc = new Scanner(System.in);
9
10         System.out.print(s: "Enter a number (1-255): ");
11         int number = sc.nextInt();
12
13         String binary = Integer.toBinaryString(number);
14
15         binary = String.format(format: "%8s", binary).replace(oldChar: ' ', newChar: '0');
16
17         System.out.println("Binary Representation: " + binary);
18
19         sc.close();
20     }
```

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```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> javac BinaryRepresentation.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> java BinaryRepresentation
Enter a number (1-255): 240
Binary Representation: 11110000
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> java BinaryRepresentation
Enter a number (1-255): 10
Binary Representation: 00001010
```

Q73. Create an Employee class with properties of your choice. Create an object of this class and also create a clone of the same. After making the clone, change the properties of the original employee object and print the properties of both the original and clone objects. Note down your observation.

Code:

Employee.java:

```
public class Employee implements Cloneable {

    private int empId;
    private String name;
    private double salary;

    // Constructor
    public Employee(int empId, String name, double salary) {
        this.empId = empId;
        this.name = name;
        this.salary = salary;
    }

    // Setter Methods
    public void setName(String name) {
        this.name = name;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }

    // Display Method
    public void display() {
        System.out.println("Employee ID : " + empId);
        System.out.println("Name      : " + name);
        System.out.println("Salary   : " + salary);
    }

    // Clone Method
    @Override
    protected Object clone() throws CloneNotSupportedException {
        return super.clone();
    }

    public static void main(String[] args) {

        try {

            // Original Object
            Employee emp1 = new Employee(101, "Arka", 50000);

            // Clone Object
```

```
Employee emp2 = (Employee) emp1.clone();

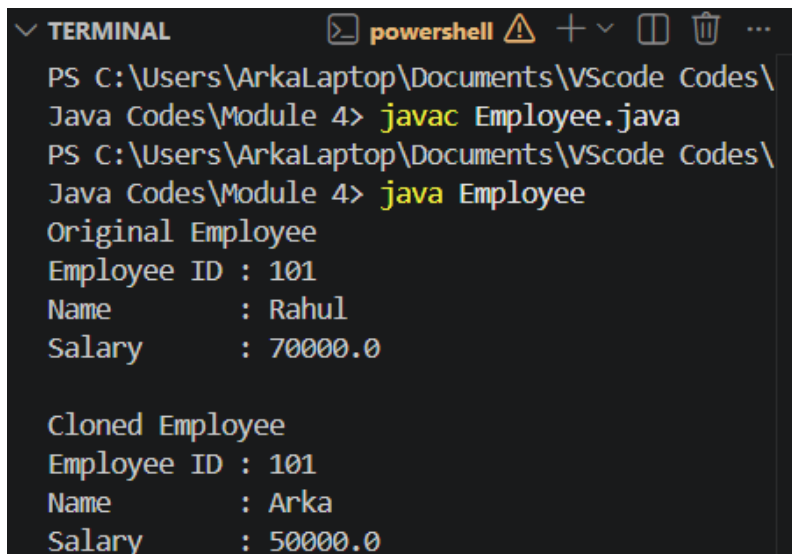
// Modify Original Object
emp1.setName("Rahul");
emp1.setSalary(70000);

System.out.println("Original Employee");
emp1.display();

System.out.println("\nCloned Employee");
emp2.display();

} catch (CloneNotSupportedException e) {
    System.out.println(e);
}
}
```

Output:



```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> javac Employee.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java Codes\Module 4> java Employee
Original Employee
Employee ID : 101
Name       : Rahul
Salary     : 70000.0

Cloned Employee
Employee ID : 101
Name       : Arka
Salary     : 50000.0
```